

Adaptive Learning versus Static Robustness: Benchmarking Reinforcement Learning and Neuroevolution for Swarm Control

Gonalo Pombo^a, Lu s Nunes^a

^a*Instituto Universit rio de Lisboa (ISCTE-IUL), ISTAR, Lisbon, Portugal*

Abstract

The control of robotic swarms is approached by two communities that rarely meet on a common benchmark: multi-agent reinforcement learning (MARL), which promises *online* adaptability, and bio-inspired optimisation, which offers pre-programmed robustness. This work provides a controlled, paradigm-agnostic comparison of three controllers — a neuroevolved graph network with attention over neighbours, Proximal Policy Optimisation (PPO) and Soft Actor–Critic (SAC) — on the same high-fidelity 3D foraging simulator with six scenarios of increasing difficulty, under deterministic evaluation and statistical testing. On task completion, gradient methods dominate: SAC solves all six scenarios and PPO five of six, failing the U-maze through *reward hacking*, whereas the evolutionary controller collapses in the bottleneck scenarios. The picture inverts under zero-shot scalability: only the attention-based controller transfers from $N = 10$ to $N = 100$ agents without retraining, since the fixed-size policies of PPO/SAC are structurally incompatible with $N \neq 20$. The hypothesis that adaptive intelligence dominates static robustness is therefore only *partially* confirmed, revealing a central trade-off between task performance at a fixed swarm size and scalability to unseen sizes.

Keywords: swarm robotics, multi-agent reinforcement learning, neuroevolution, zero-shot scalability, graph neural networks, benchmarking

1. Introduction

Robotic swarms promise scalable and resilient solutions for tasks such as search-and-rescue, environmental monitoring and distributed logistics. The central engineering question is which control paradigm best sustains operability under uncertainty. Two schools dominate. *Bio-inspired optimisation* (e.g. evolutionary algorithms, particle swarm optimisation) searches a global parameter space *offline* for robust, static controllers (Kennedy and Eberhart, 1995). *Multi-agent reinforcement learning* (MARL) instead learns adaptive policies *online* through interaction (Huh and Mohapatra, 2023). Despite extensive work on each, direct and rigorous comparisons across paradigms remain scarce (Majid et al., 2024; Bettini et al., 2024): studies typically refine a single algorithm against variants of its own family, neglecting cross-paradigm baselines.

This paper addresses that gap with an impartial benchmark. We implement a high-fidelity 3D foraging simulator and evaluate three controllers — a neuroevolved graph network with self-attention over neighbours, PPO and SAC — across six scenarios, along three axes: task performance (P_{task}), robustness to agent failures (R_{robust}) and zero-shot scalability (S_{scale}), all supported by non-parametric significance testing. Our contribution is twofold: (i) a reproducible, paradigm-agnostic evaluation protocol; and (ii) evidence of a central trade-off, in which the gradient methods win on task completion at a fixed swarm size while only the graph-based controller scales to unseen swarm

sizes. The full methodology and results are detailed in the accompanying dissertation.

2. Related Work

Deep reinforcement learning has been applied to swarm systems with parameter sharing to tame the joint-action explosion (H ttenrauch et al., 2019), and graph neural networks (GNNs) have emerged as the key enabler of scalability: by modelling the swarm as a graph and attending over neighbours, a policy trained with few agents can be deployed with many (Chen et al., 2024). On the bio-inspired side, neuroevolution optimises network weights without gradients (Salimans et al., 2017), offering stable controllers for static tasks but limited *online* adaptability (Kegeleirs and Birattari, 2025). Comparative surveys of deep RL versus evolution strategies highlight the lack of a common performance reference across the two families (Majid et al., 2024), while standardised MARL benchmarks remain focused within the learning community (Bettini et al., 2024). Applications of deep RL to swarm navigation (Iskandar et al., 2023) illustrate the type of single-paradigm study that this work extends with a systematic, statistically grounded comparison against the bio-inspired alternative.

3. Methodology

3.1. Simulation environment

The environment is a 3D continuous foraging task in a circular arena (radius 15 m) with $N = 20$ homogeneous agents

Email address: goncalopombo123@gmail.com (Gonalo Pombo)

that must collect food and return it to a nest. Six scenarios of increasing difficulty are used: an open *Sandbox*; three navigation mazes (*U-wall*, *Bottleneck*, *Four Rooms*); and two cooperative tasks (*Cooperative Door*, requiring ≥ 3 agents to open a passage, and *Cooperative Perception*, with a moving target). Each agent observes an egocentric homing compass to the nest, eight LiDAR rays for local obstacle sensing, and the relative state of neighbours — a partially observable setting. To avoid local minima in the maze scenarios, the progress reward uses a *geodesic* potential computed by Dijkstra’s algorithm over an inflated occupancy grid, acting as potential-based reward shaping (Ng et al., 1999) that does not alter the optimal policy and is used as a training signal only, never as an observation.

3.2. Controllers

The *evolutionary controller* (GNN) is a compact network that encodes the egocentric features and applies scaled dot-product self-attention over the neighbour set (Vaswani et al., 2017), producing an action invariant to the number of neighbours. Its weights are optimised by a $(1+\lambda)$ evolution strategy (population 30, truncation with elitism, Gaussian mutation), with a fitness that prioritises the task, $J = \bar{f} \cdot 10^4 + 5000 \tanh(\bar{R}/5000)$, where $\bar{f}$ is the mean food collected and $\bar{R}$ the mean shaped reward; the tanh term prevents the saturation that would otherwise make selection blind. The *learning baselines* are PPO (Schulman et al., 2017) and SAC (Haarnoja et al., 2018) from Stable-Baselines3, using a shared multilayer perceptron policy ([256, 256]) over the dense neighbourhood observation (parameter sharing). The attention-over-graph mechanism is thus instantiated only in the evolutionary controller, which is what enables its zero-shot transfer.

3.3. Evaluation protocol

All controllers are evaluated deterministically over 30 paired-seed episodes per scenario. We report P_{task} (success rate, with success defined as at least one item collected) and the mean food collected per episode, a metric comparable across paradigms. Robustness (R_{robust}) injects the failure of 10% of agents at mid-episode; scalability (S_{scale}) tests $N \in \{10, 20, 50, 100\}$ without retraining. Pairwise differences are assessed with the Mann–Whitney U test, complemented by Welch’s t -test, at $\alpha = 0.05$.

4. Results

4.1. Task performance

Table 1 reports the food collected per episode — the magnitude metric — alongside the success rate. We stress the former: a success rate of 100% under the definition “at least one item collected” is a weak signal, as the table makes clear. The gradient methods dominate the task: SAC collects substantially in all six scenarios and PPO in five of six, failing only the *U-wall*, where it collects *zero* items despite accumulating a large shaped reward — a clear case of *reward hacking*, in which the exploration bonus is exploited without foraging. The evolutionary

controller is the weakest by a wide margin: although it occasionally reaches a non-zero success rate (e.g. 97% in Cooperative Perception), it collects only residual amounts (2.6 items there, and essentially zero in the three mazes). This gap between a high success rate and a near-zero yield is the signature of *fitness exploitation*: the training fitness rises without translating into task completion. Of the 18 pairwise comparisons on food collected, 17 are significant ($p < 0.001$); the only exception is PPO vs. SAC in Four Rooms.

4.2. Zero-shot scalability

The trade-off emerges under scalability. Only the attention-based controller is viable for $N \neq 20$: its success rate *increases* with swarm size, from 15% at $N = 10$ to 100% at $N = 100$, with no per-capita degradation. The MLP policies of PPO/SAC have a fixed-size input and are structurally incompatible with other swarm sizes. Architectural adaptability — the property the hypothesis attributes to MARL — thus manifests in the *representation* (graph with attention), not in the optimisation algorithm.

4.3. Robustness

Under the failure of 10% of agents at mid-episode, all paradigms retain high performance (94–99% of collected items for PPO/SAC; 69–96% for the GNN where it functions), with no collapse: parameter sharing and local observation confer functional redundancy across the swarm.

5. Discussion

The collapse of the evolutionary controller in the maze scenarios is not a property of the architecture — the same graph-with-attention representation underpins its scalability advantage — but of the interaction between the optimisation method and the reward structure. Neuroevolution optimises from a single scalar fitness per episode, without temporal credit assignment; it never learns *which* action mattered. When the task reward is sparse, as in the mazes, almost every genome collects no food, fitness reduces to the (saturated) shaping term, and the population settles on a *fitness plateau* where selective pressure vanishes. The same stochastic nature explains the high variance between independent runs, absent in the gradient methods: without a gradient pulling every run to a common optimum, each run is a different random walk over the fitness landscape, and finding the narrow valley of maze solutions depends largely on the number of generations explored.

Two implications follow. First, the comparison combines two factors — optimisation algorithm and network architecture — so the limitation observed is one of *sampling efficiency* of the optimiser, not representational capacity; integrating the attention module into a gradient-trained policy is the natural next step to isolate the effect. Second, the engineering value is not in electing a winner but in identifying the inflection point: the choice of paradigm should follow the dominant axis of demand — task performance at a fixed swarm size (MARL) versus scalability to unseen sizes (graph with attention).

Table 1: Per-episode performance under deterministic evaluation (30 paired-seed episodes per scenario, official run): mean food collected $\pm$ standard deviation, with the task success rate (percentage of episodes with at least one item collected) in parentheses. Food collected is the magnitude metric compared across paradigms; the success rate alone is a weak indicator, since an agent reaches 100% by collecting a single item. Best mean per row in bold.

Scenario	GNN	PPO	SAC
Sandbox	0.8 $\pm$ 0.7 (67%)	73.3 $\pm$ 5.1 (100%)	25.2 $\pm$ 3.6 (100%)
U-wall	0.4 $\pm$ 0.5 (43%)	0.0 $\pm$ 0.0 (0%)	20.6 $\pm$ 1.3 (100%)
Bottleneck	0.0 $\pm$ 0.0 (0%)	41.0 $\pm$ 1.4 (100%)	35.0 $\pm$ 1.6 (100%)
Four Rooms	0.0 $\pm$ 0.0 (0%)	11.5 $\pm$ 3.4 (100%)	11.6 $\pm$ 3.0 (100%)
Cooperative Door	0.0 $\pm$ 0.0 (0%)	67.7 $\pm$ 1.6 (100%)	60.5 $\pm$ 1.3 (100%)
Cooperative Perception	2.6 $\pm$ 1.4 (97%)	17.9 $\pm$ 1.9 (100%)	16.2 $\pm$ 1.8 (100%)

6. Conclusions

We presented an impartial benchmark of a neuroevolved attention controller against PPO and SAC on a 3D foraging swarm, across six scenarios and three axes of evaluation with statistical support. The hypothesis that adaptive intelligence dominates static robustness is only partially confirmed: gradient methods win decisively on task completion at $N = 20$, while only the attention-based controller scales zero-shot to $N = 100$. Future work includes a gradient-trained attention policy to disentangle optimiser from architecture, a multi-run protocol to consolidate variance estimates, and validation of the sim-to-real transfer (Tobin et al., 2017).

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